PROJECT 1 – HOUSE PRICE PREDICTION IN INDIA

Regression Analysis using the Bengaluru House Price Dataset

1. Objective

The objective of this project is to develop a regression model to predict house prices using property characteristics such as area, location, BHK and number of bathrooms. The project follows the required stages of data exploration, cleaning, EDA, feature engineering, model building, evaluation and interpretation.

2. Dataset & Preprocessing

The original Bengaluru housing dataset contained **13,320 records and 9 columns**. The target variable was **price**. Missing values were identified in location, size, society, bath and balcony. The society column was removed because of its high number of missing observations. The size field was converted into a numerical **BHK** feature, and total_sqft was converted from text/range values into numerical values using the midpoint for ranges. Missing bath and balcony values were filled with their medians.

Properties below 300 sqft and properties with less than 300 sqft per BHK were removed as consistency filters. Two engineered variables, **sqft_per_bhk** and **price_per_sqft**, were created. The final cleaned dataset contained **12,513 rows and 10 columns**.

3. Exploratory Data Analysis

EDA included price and area distributions, price versus total area, price versus BHK, price versus bathrooms, location-based analysis, price-per-square-foot analysis, area versus BHK, and a correlation heatmap. The analysis indicated that total property area is strongly associated with house price. Location also showed substantial variation in property values, while BHK and bathroom count provided additional information about property size and configuration.

4. Machine Learning Models

The data was divided into training and testing sets using an 80:20 split. Categorical variables were converted using one-hot encoding. Three regression models required by the project were trained: Linear Regression, Decision Tree Regression and Random Forest Regression. Price_per_sqft was excluded from model inputs because it is directly derived from the target price and would cause target leakage.

Model	MAE	RMSE	R^2
Linear Regression	42.54	96.13	0.5545
Decision Tree	37.76	121.66	0.2865
Random Forest	**32.00**	**86.94**	**0.6356**

5. Results & Interpretation

Random Forest performed best, achieving the lowest MAE (31.996) and RMSE (86.945) and the highest R^2 (0.6356). Therefore, it was selected as the best-performing model among the three tested algorithms. Its R^2 indicates that the model explains approximately **63.56%** of the variation in house prices in the test data. Decision Tree produced a lower MAE than Linear Regression but a much higher RMSE and lower R^2, indicating larger prediction errors on some observations.

6. Top 5 Features

Rank	Feature	Importance
1	total_sqft	64.15%
2	location – Defence Colony	2.98%
3	bath	2.92%
4	bhk	2.40%
5	availability – Ready To Move	2.39%

Total square footage was the dominant feature in the Random Forest model, with an importance of approximately 64.15%. Location, bathroom count, BHK and availability also contributed to the model's predictions. The location and availability entries above are individual one-hot encoded categories.

7. Conclusion & Business Insight

The project demonstrates an end-to-end machine-learning approach to Bengaluru house-price prediction. Among the evaluated models, Random Forest provided the strongest performance. The results suggest that property area is the most important predictive factor, while location and property configuration provide additional information. A system based on this model could support

preliminary property valuation, listing comparison and market analysis. It should be treated as a decision-support tool rather than a definitive valuation.

Project status: Project 1 completed according to the regression tasks specified in the Major Project brief. ■filecite■turn0file0■L4-L31■